

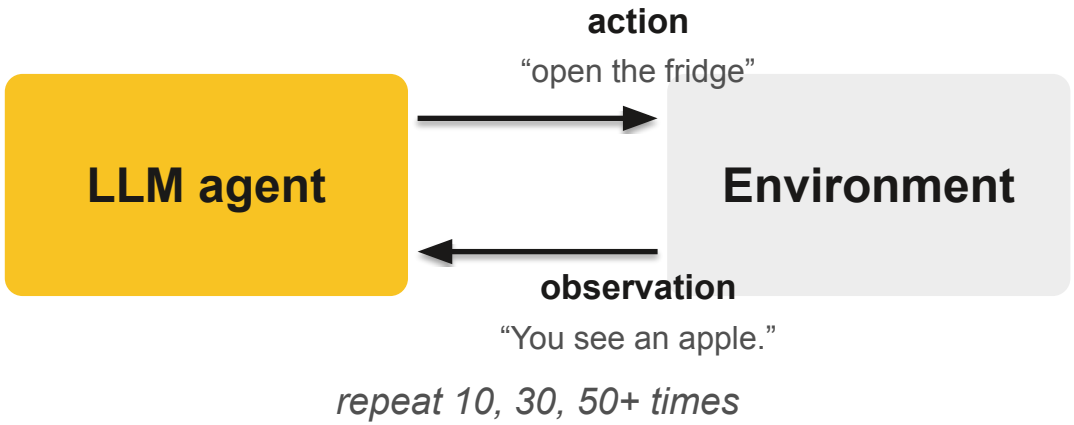
Improving Credit Assignment for Long Horizon Agent Reinforcement Learning

CAP 6938: AI Knowledge Systems Term Project

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Motivation

Agents now act for dozens of steps before anyone knows whether they succeeded.



One episode



No feedback on any single step.
Only the final outcome is scored.

Coding agents

edit files, run tests, read errors, try again

Computer & web agents

click, type and navigate across many pages

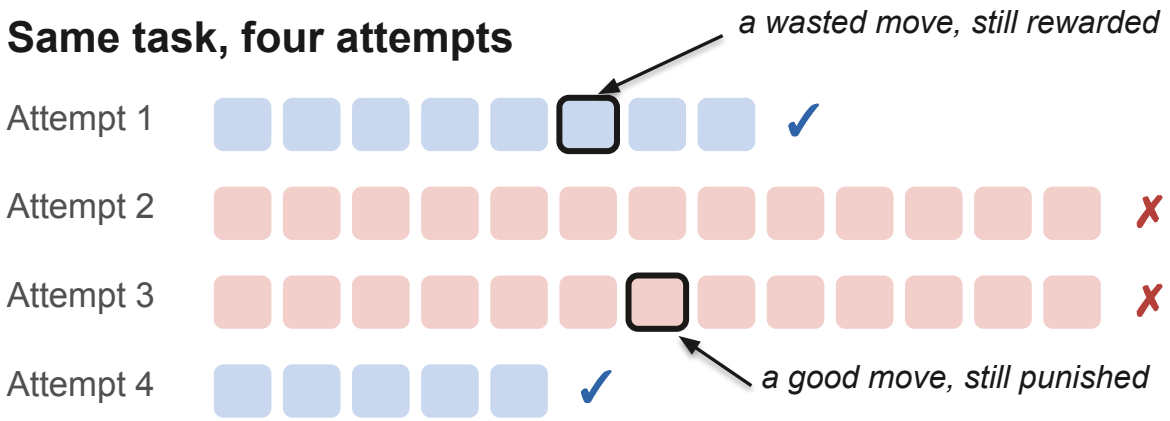
Household robots

find, pick up, clean and place objects

Motivation: Credit Assignment

Critic-free RL (GRPO and its successors) scores a whole attempt with one number.

Same task, four attempts



advantage = (reward − group average) / group std

No value network: attempts are compared with each other. Every step of an attempt gets the same score.

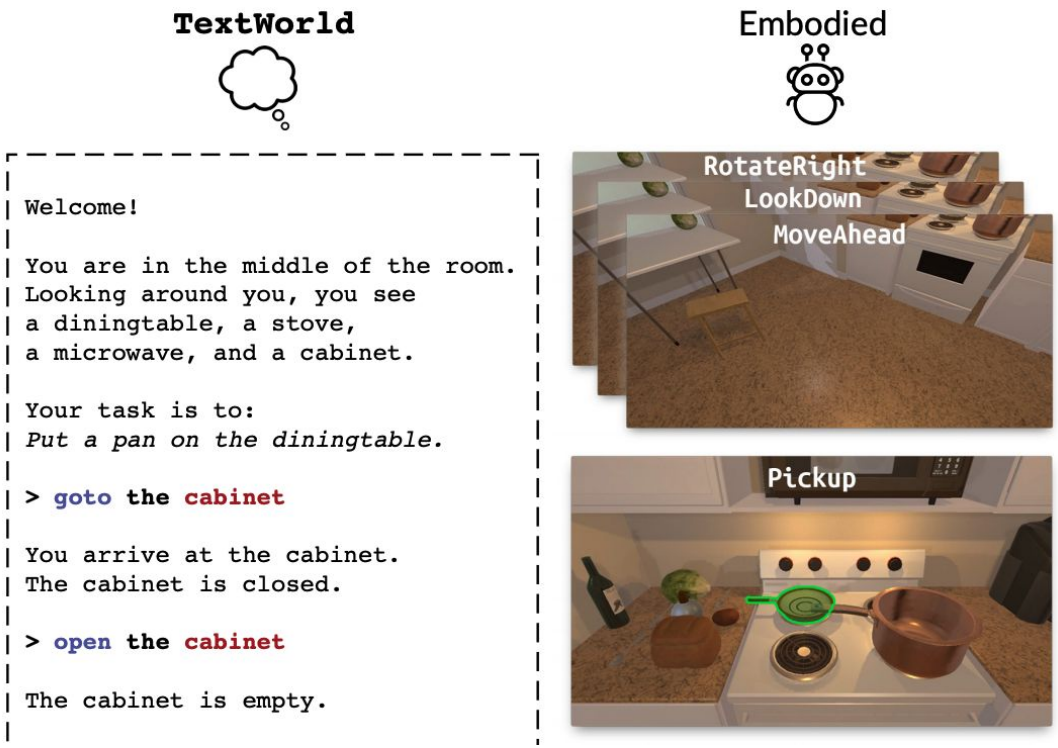
Zoom into one step

```
<think> The ladle is clean. The only cabinet
described is cabinet 1, which is open. I
should put it there. </think>
<action> move ladle 2 to cabinet 1 </action>
```

About 100 tokens. Step-level methods (GiGPO, GACA) score each step, but every token inside a step still gets the same credit.

Research Question: which tokens actually made the decision, and can we assign the credit there?

Benchmark



ALFWorld pairs a text game (left) with a 3D household simulator (right). We train and test on the text version. Image: Shridhar et al., ICLR 2021 (MIT license).

Training how the model learns

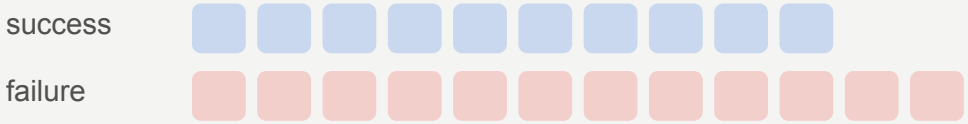
- **3,553** training games, **6** task types
- Each iteration: **4 games × 8 attempts**, temperature 1.0
- **34 iterations** = 1,088 episodes (about 136 games)
- Up to 50 actions; reward 10 only if solved

Evaluation how we score it

- **128 held-out games**, never used in training
- The **same 128 games** for every model
- **1 attempt** per game, temperature 0.4, up to 50 actions
- Score = % of games solved, at the **final iteration (34)**

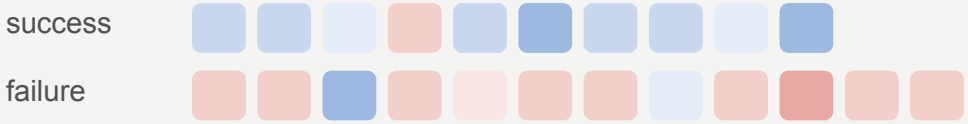
Baselines

GRPO one score per attempt



Run the task 8 times. Each attempt is scored against the group average, and every step gets that score.

GACA adds a per-step score



When attempts reach the same situation, compare what they did next (step score). A gate leans on the step score where the model was unsure, and on the attempt score elsewhere.

Our baseline results

Model	Success	Solved
Qwen2.5-1.5B (untrained)	4.7%	6 / 128
+ GRPO	14.1%	18 / 128
+ GACA	**18.8%**	**24 / 128**

Final checkpoint after 34 training iterations (lr 1e-6), on the same 128 held-out games. GACA solves 24 games, GRPO 18.

Experiments

COMMIT: give the credit to the tokens where the agent makes up its mind.

1 Cut

Split the reasoning every 16 tokens.

2 Probe

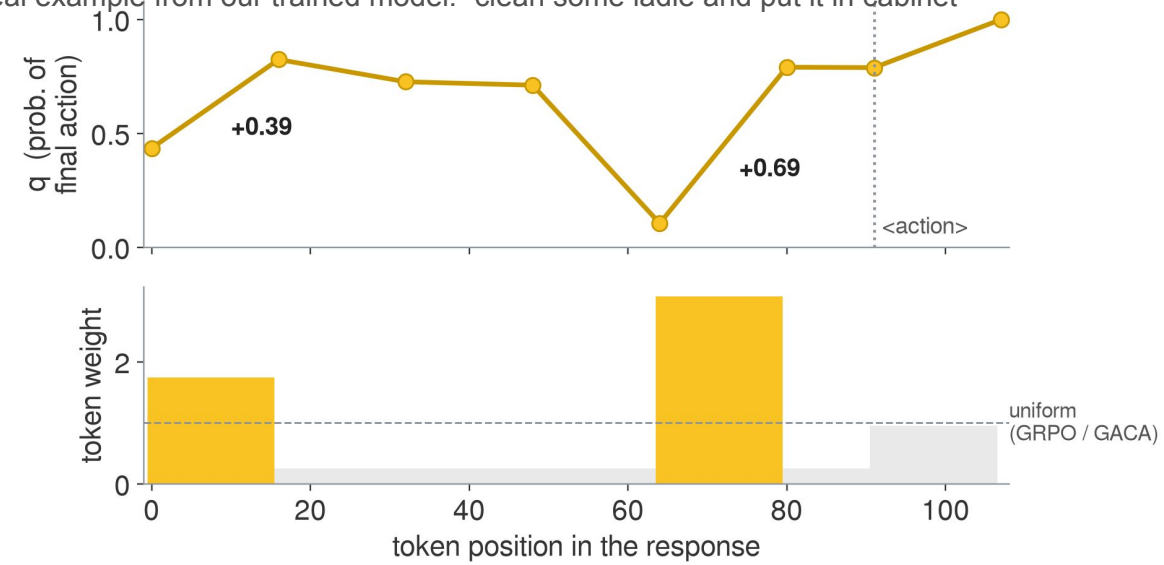
At each cut, make the model answer right away. q = the chance it picks the action it finally took.

3 Weight

Where q jumps, tokens get more weight. Token credit = GACA step score × weight.

No critic network. Weights average 1, so each step keeps its total credit.

Real example from our trained model: “clean some ladle and put it in cabinet”



“…you need to clean the ladle and”

q: 0.43 → 0.82

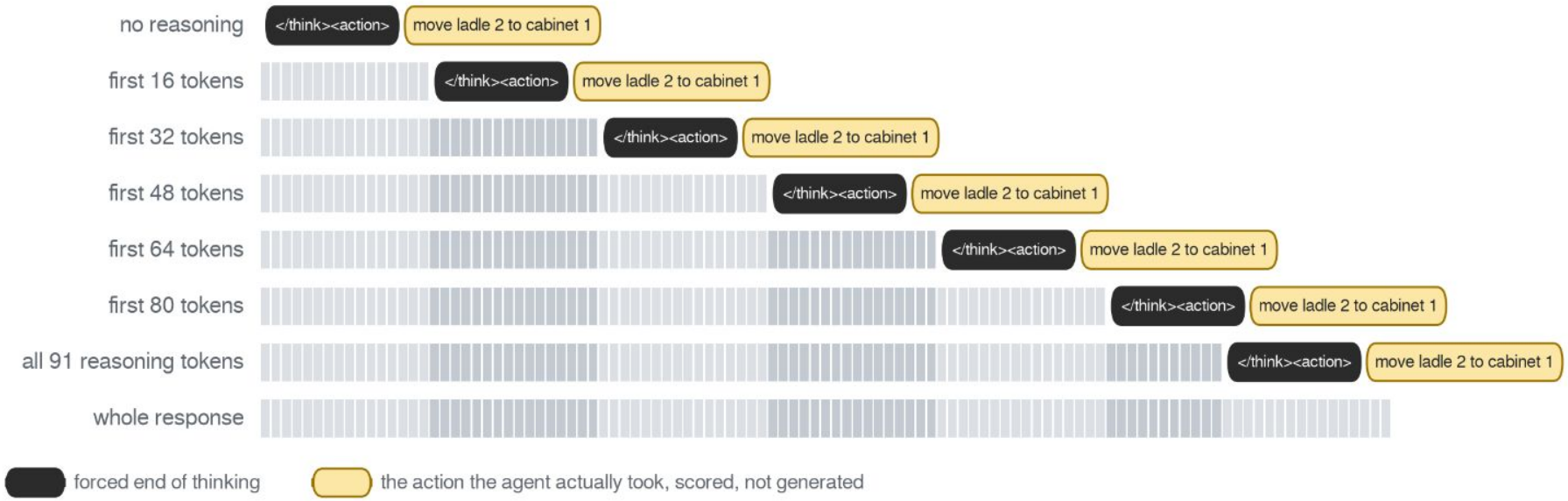
“You should clean the ladle, put it into this cabinet”

q: 0.10 → 0.79

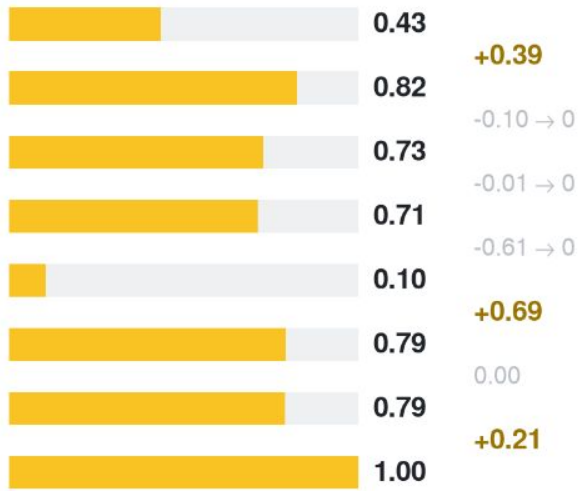
How the COMMIT probe decides which tokens get the credit

One real step from our trained agent · task: "clean some ladle and put it in cabinet" · 107 response tokens, cut every 16

1 Re-read the finished response at growing lengths

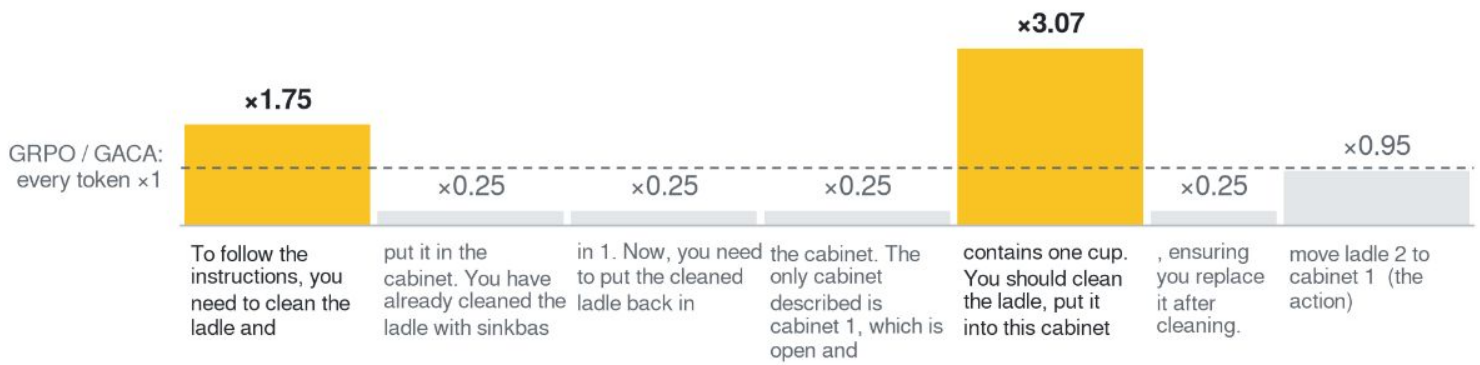


2 Probability of the real action



one forward pass per row · gold numbers = rise

3 Each 16-token chunk gets credit in proportion to how much it raised the probability



Token advantage

$$A_{\text{token}} = A_{\text{GACA}} \times w_{\text{token}}$$

This step: A_GACA = +2.84

"contains one cup..." tokens: 2.84 × 3.07 = 8.73

"...which is open and" tokens: 2.84 × 0.25 = 0.71

Chunks where the probability fell or stayed flat drop to the floor (×0.25). Weights are clipped to [0.25, 4] and average 1 per step, so the step's total credit is unchanged; only where it lands moves.

Experiment Setup

Same recipe as the GACA paper, at about 3% of its training budget.

	Ours	GACA paper
Model	Qwen2.5-1.5B-Instruct	same
Training framework	verl-agent	same
Tasks × attempts per iteration	4 × 8	16 × 8
Training iterations	34	300
Seeds per method	1	1
Episodes seen in training	1,088	38,400

Evaluation: 128 held-out games, one attempt each, final checkpoint only. Each method differs only in how it assigns credit.

1 GPU rented A100 40 GB, $0.59 / hour

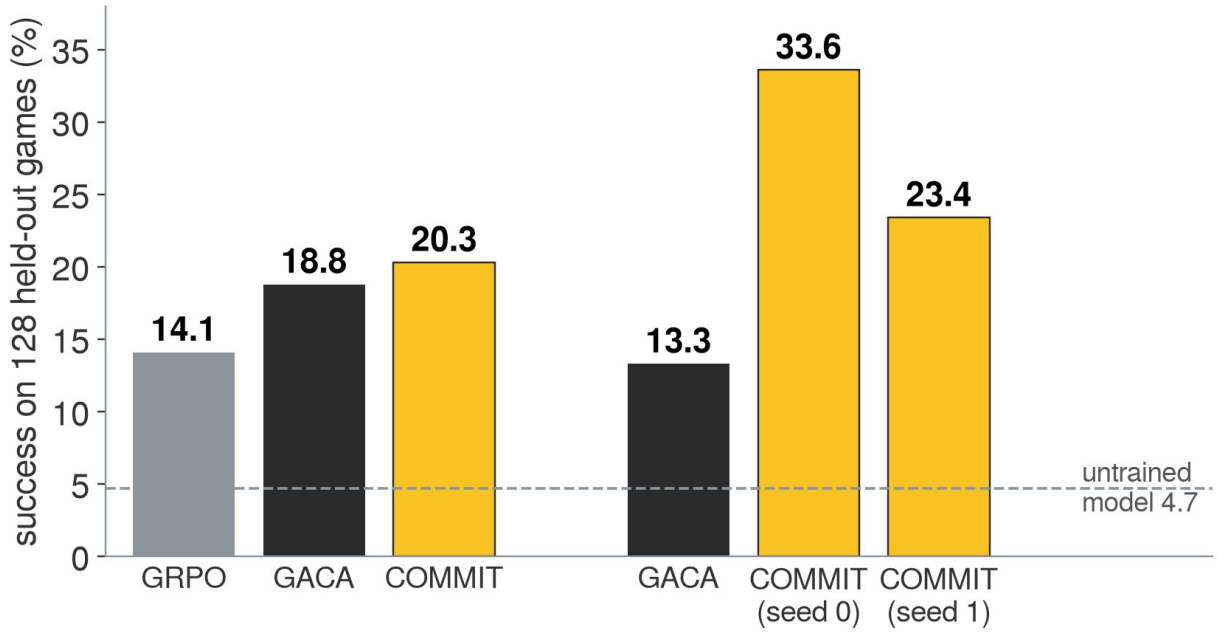
4–5 h per training run

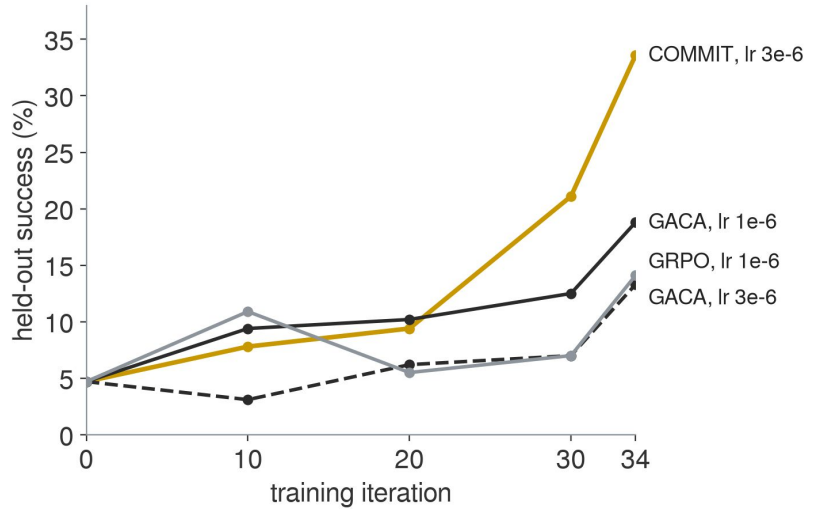
~70 GPU-h for our 16 training runs

At paper scale: ~2,400 GPU-hours
≈ $1,400 and about 3 months on our one GPU, for our 16 configurations.

Benchmark Results

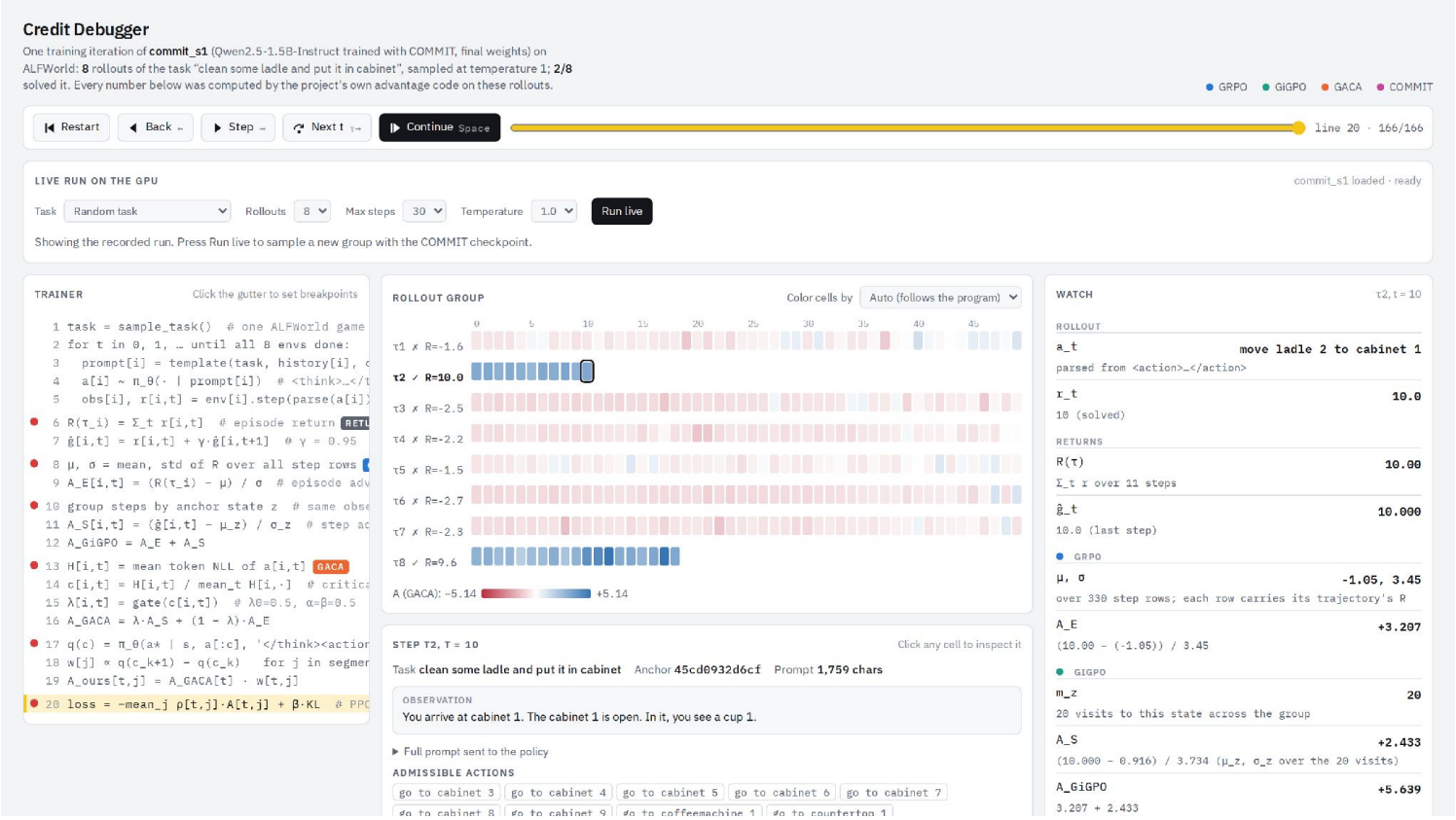
Same settings, same seed: COMMIT solved 43 of 128 games. Best GACA solved 24.





At lr 3e-6, COMMIT solved 43 of 128 games and GACA 17. A second COMMIT run, on another seed, reached 23.4% and also beat GACA.

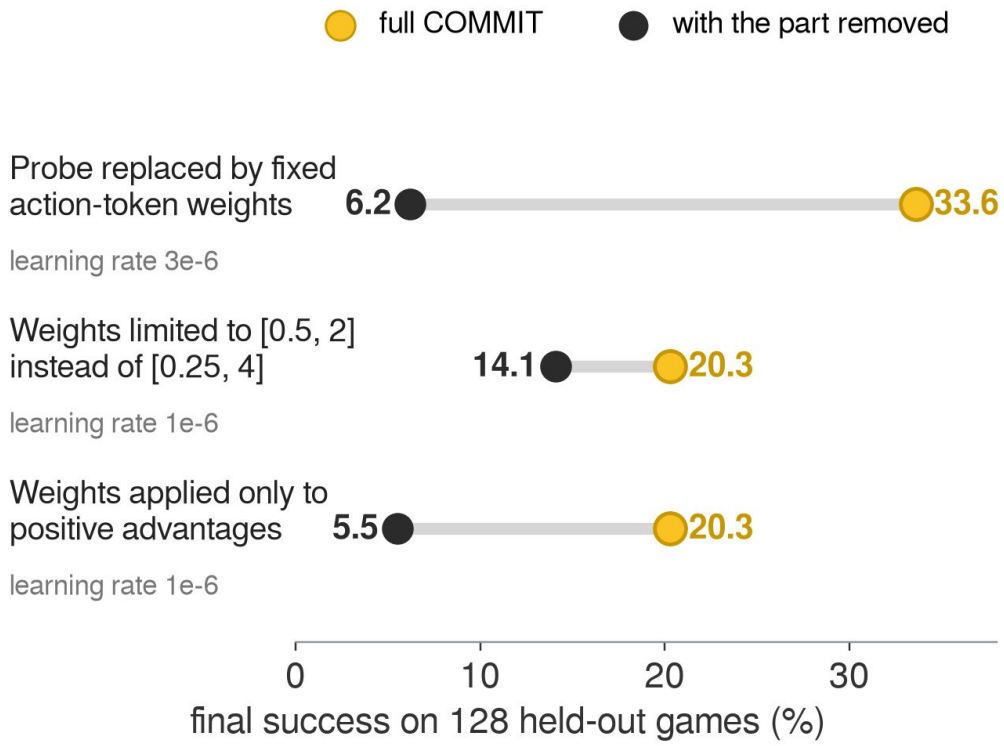
Live Demo

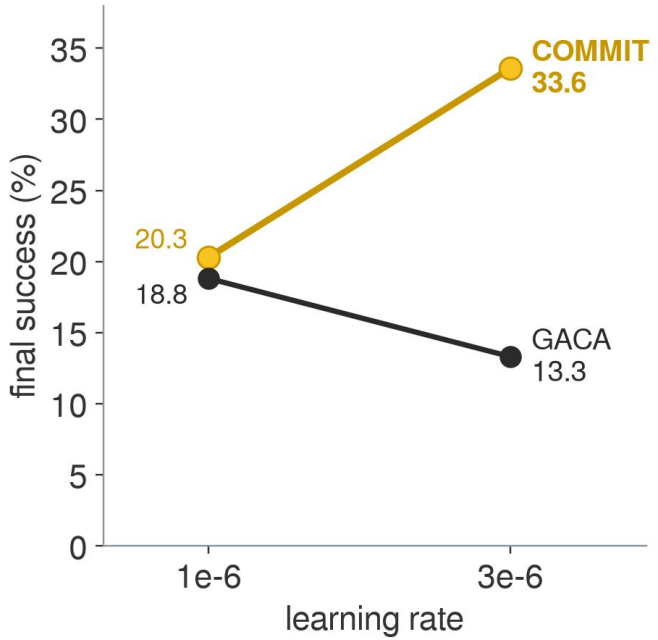


Credit Debugger: replays one training iteration line by line, or samples a new one live on the GPU.

Ablation Study

Take away one part of COMMIT and performance drops.





Larger steps help COMMIT and hurt GACA.

Conclusions

What our data supports

- COMMIT beat GACA at lr 3e-6: 43 vs 17 games, and won on a second seed too
- The probe is what matters: a fixed action-token rule collapsed at the same setting (6.2%)
- Weights must be sharp and two-sided: flatter (14.1%) or positive-only (5.5%) weights hurt
- COMMIT gets better with a larger learning rate; GACA gets worse.

Likely, not yet proven

- Probe weights make the gradient less noisy (about 3× lower in our measurement), which is why bigger steps help
- Under a stated assumption, the probe weight equals the exact credit each token deserves (we proved this)
- Too large a step still breaks training; a probe that drifts with the policy may be the reason

Open questions

- Does the lead hold at the paper's 300 updates?
- Why does the fixed action-token rule collapse?
- Would negative credit where commitment drops work better?
- Does it carry over to larger models and other agent tasks?

Future Work

Episodes seen per training run

Ours: 1,088

GACA paper: 38,400

With more compute

- Train the full 300 updates with 16 tasks each
- Several seeds per method
- Larger models, e.g. Qwen2.5-7B
- More agent benchmarks: WebShop, coding and web agents

Most promising experiments

- Signed credit: negative weight where commitment drops
- Probe scored by the frozen starting model, so the weights cannot drift
- Compare against a learned critic (PPO)



Thank You!

Questions?

References

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